Exponential Smoothing

Complete Easy Method + Line-by-Line Calculation

Problem: Calculate exponential smoothing for the given demand data using α **= 0.25**.

1. What is Exponential Smoothing?

Exponential smoothing is a forecasting method used to estimate a new value from the current actual value and the previous forecast/smoothed value.

In very easy words: **take a part of the current actual value + take the remaining part of the previous forecast = new forecast.**

2. Why do we use it?

• It helps forecast future demand or sales.

• It reduces sudden ups and downs in the data.

• It gives a smoother series than the raw actual data.

• It is commonly used in sales, demand, inventory, and time-series forecasting.

3. Important Terms

Term	Easy meaning
Actual (A)	The real value observed in a period.
Forecast (F)	The estimated/smoothed value.
Alpha (α)	Weight given to the current actual value.
$1 - \alpha$	Weight given to the previous forecast.

4. Formula

General formula:

$$F_t = \alpha A_t + (1 - \alpha)F_{t-1}$$

For this problem, $\alpha = 0.25$:

$$F_t = (A_t \times 0.25) + (F_{t-1} \times 0.75)$$

Why 0.75? Because: 1 – 0.25 = 0.75.

5. What does α = 0.25 mean?

α = 0.25 means the current actual value receives **25%** weight, while the previous forecast receives **75%** weight.

Current Actual = 25% weight
Previous Forecast = 75% weight

Because α is relatively low, the result changes more slowly and the smoothing is stronger.

6. Given Data

Period	Actual Demand	Forecast (α = 0.25)
1	120	120.0000
2	125	121.2500
3	123	121.6875
4	130	123.7656
5	128	124.8242
6	135	127.3682
7	140	130.5261
8	138	132.3946
9	145	135.5459
10	150	139.1595
11	148	141.3696
12	155	144.7772
13	160	148.5829
14	158	150.9372
15	165	154.4529
16	170	158.3397
17	168	160.7547
18	175	164.3161
19	180	168.2370
20	178	170.6778

7. Step 1 — Initial Forecast

For the first period, we start with the first actual value:

$$F_1 = A_1 = 120$$

Therefore, F_1 = 120.

8. Easy Calculation Rule for Every Next Period

Always repeat these three steps:

Step 1: Current Actual × 0.25
Step 2: Previous Forecast × 0.75
Step 3: Add Step 1 + Step 2

That answer becomes the forecast for the current period.

9. Detailed Calculations — Periods 2 to 4

Period 2
Current Actual = 125
Previous Forecast = 120.0000000000
Formula: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)
= (125 × 0.25) + (120.0000000000 × 0.75)
= 31.2500000000 + 90.0000000000
= **121.2500000000**

Period 3
Current Actual = 123
Previous Forecast = 121.2500000000
Formula: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)
= (123 × 0.25) + (121.2500000000 × 0.75)
= 30.7500000000 + 90.9375000000
= **121.6875000000**

Period 4
Current Actual = 130
Previous Forecast = 121.6875000000
Formula: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)
= (130 × 0.25) + (121.6875000000 × 0.75)
= 32.5000000000 + 91.2656250000
= **123.7656250000**

9. Detailed Calculations — Periods 5 to 7

Period 5
Current Actual = 128
Previous Forecast = 123.7656250000
Formula: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)
= (128 × 0.25) + (123.7656250000 × 0.75)
= 32.0000000000 + 92.8242187500
= **124.8242187500**

Period 6
Current Actual = 135
Previous Forecast = 124.8242187500
Formula: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)
= (135 × 0.25) + (124.8242187500 × 0.75)
= 33.7500000000 + 93.6181640625
= **127.3681640625**

Period 7
Current Actual = 140
Previous Forecast = 127.3681640625
Formula: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)
= (140 × 0.25) + (127.3681640625 × 0.75)
= 35.0000000000 + 95.5261230469
= **130.5261230469**

9. Detailed Calculations — Periods 8 to 10

Period 8
Current Actual = 138
Previous Forecast = 130.5261230469
Formula: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)
= (138 × 0.25) + (130.5261230469 × 0.75)
= 34.5000000000 + 97.8945922852
= **132.3945922852**

Period 9
Current Actual = 145
Previous Forecast = 132.3945922852
Formula: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)
= (145 × 0.25) + (132.3945922852 × 0.75)
= 36.2500000000 + 99.2959442139
= **135.5459442139**

Period 10
Current Actual = 150
Previous Forecast = 135.5459442139
Formula: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)
= (150 × 0.25) + (135.5459442139 × 0.75)
= 37.5000000000 + 101.6594581604
= **139.1594581604**

9. Detailed Calculations — Periods 11 to 13

Period 11
Current Actual = 148
Previous Forecast = 139.1594581604
Formula: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)
= (148 × 0.25) + (139.1594581604 × 0.75)
= 37.0000000000 + 104.3695936203
= **141.3695936203**

Period 12
Current Actual = 155
Previous Forecast = 141.3695936203
Formula: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)
= (155 × 0.25) + (141.3695936203 × 0.75)
= 38.7500000000 + 106.0271952152
= **144.7771952152**

Period 13
Current Actual = 160
Previous Forecast = 144.7771952152
Formula: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)
= (160 × 0.25) + (144.7771952152 × 0.75)
= 40.0000000000 + 108.5828964114
= **148.5828964114**

9. Detailed Calculations — Periods 14 to 16

Period 14
Current Actual = 158
Previous Forecast = 148.5828964114
Formula: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)
= (158 × 0.25) + (148.5828964114 × 0.75)
= 39.5000000000 + 111.4371723086
= **150.9371723086**

Period 15
Current Actual = 165
Previous Forecast = 150.9371723086
Formula: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)
= (165 × 0.25) + (150.9371723086 × 0.75)
= 41.2500000000 + 113.2028792314
= **154.4528792314**

Period 16
Current Actual = 170
Previous Forecast = 154.4528792314
Formula: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)
= (170 × 0.25) + (154.4528792314 × 0.75)
= 42.5000000000 + 115.8396594236
= **158.3396594236**

9. Detailed Calculations — Periods 17 to 19

Period 17
Current Actual = 168
Previous Forecast = 158.3396594236
Formula: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)
= (168 × 0.25) + (158.3396594236 × 0.75)
= 42.0000000000 + 118.7547445677
= **160.7547445677**

Period 18
Current Actual = 175
Previous Forecast = 160.7547445677
Formula: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)
= (175 × 0.25) + (160.7547445677 × 0.75)
= 43.7500000000 + 120.5660584258
= **164.3160584258**

Period 19
Current Actual = 180
Previous Forecast = 164.3160584258
Formula: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)
= (180 × 0.25) + (164.3160584258 × 0.75)
= 45.0000000000 + 123.2370438193
= **168.2370438193**

9. Detailed Calculations — Periods 20 to 20

Period 20
Current Actual = 178
Previous Forecast = 168.2370438193
Formula: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)
= (178 × 0.25) + (168.2370438193 × 0.75)
= 44.5000000000 + 126.1777828645
= **170.6777828645**

10. Final Answers

Period	Actual	Forecast
1	120	120.0000
2	125	121.2500
3	123	121.6875
4	130	123.7656
5	128	124.8242
6	135	127.3682
7	140	130.5261
8	138	132.3946
9	145	135.5459
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11. Important Check

The values in your original table such as **124.75, 123.0875, and 129.654375** were calculated using α = 0.95. Since you asked to use α **= 0.25**, the correct answers for this PDF are the values calculated above with 0.25 and 0.75.

12. Easy Exam Notes

Definition: Exponential smoothing is a forecasting technique that uses a weighted combination of the current actual value and the previous forecast value.

Formula: $F_t = \alpha A_t + (1 - \alpha)F_{t-1}$

For α = 0.25: New Forecast = (Actual × 0.25) + (Previous Forecast × 0.75)

Memory trick: Low α → more smoothing. High α → more importance to recent actual data.